

 HE-L1 · Inflow Layer

CO_1 HN Hotels Pvt Ltd · HE

Closed circle for Hamza Express. Identity is POS-user-PIN (flexible, any logged-in employee owns the cash). UPI flows direct to bank — no per-captain promise pile.

SOURCES

5

POS configs at test.hamzahotel.com

CASH ACTORS

PIN

flexible · POS user logged-in

DESTINATIONS

3

D1 · BNK_HDFC_HE · D3

ACTIVE PMS

12

11 + 14 + 19 + 52 + 58 + ...

UPI GATEWAY

Paytm

CSV reconcile · API key pending

AGGREGATOR SRC

scrape

/ops/aggregator extension · same as NCH

ARCHITECTURAL DELTAS VS NCH-L1 · SAME SKELETON, SIMPLER MECHANIC

IDENTITY

NCH: RNR_64–RNR_68 (5 fixed runners with barcodes + dedicated QRs)

HE: POS_USER_PIN — whoever logs into the POS to take the order owns settlement responsibility for that ticket. Flexible, anyone can be on shift.

CAPTAIN PROMISE

NCH runner: owes cash AND UPI (per-runner Razorpay QR fan-in)

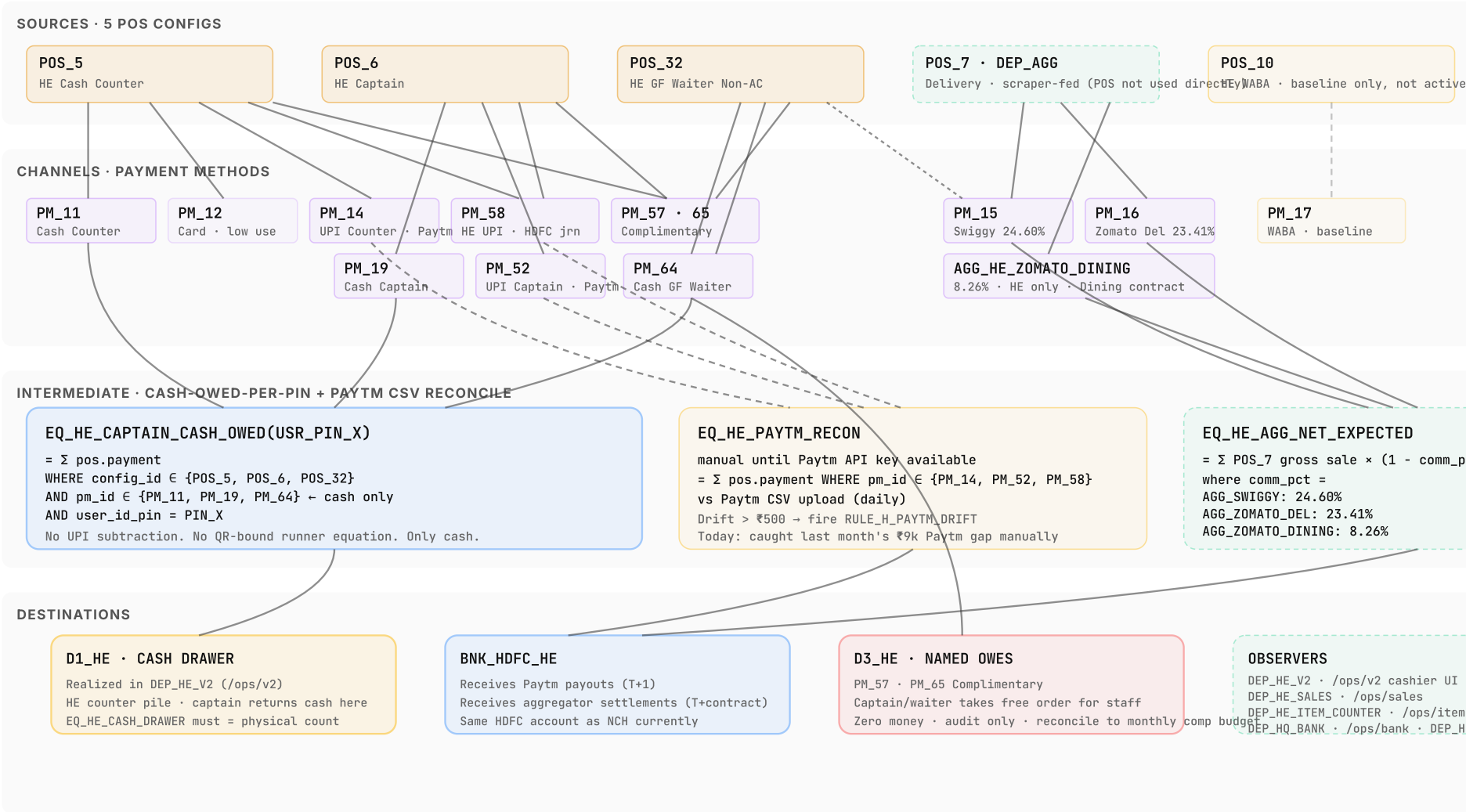
HE captain/waiter: owes **only cash**. UPI flows direct to the bank — no per-captain QR. Settlement equation is pure cash. Simpler.

UPI GATEWAY

NCH: Razorpay live API + webhook (✓ events real-time)

HE: Paytm (no API key yet) → manual CSV reconciliation by agent until API arrives. Lower volume = no live data isn't a deal-breaker.

THE CLOSED CIRCLE · POS → PM → CASH-OWED-BY-PIN → LAND



LIVE STATE FUTURE

Computed by Claude agent: `closing_inv + new_purchase - live_pos_sale = live_count`

CHICKEN KABAB RAW

—

threshold < 5kg

MUTTON RAW

—

threshold < 3kg

BIRYANI RICE RAW

—

threshold < 10kg

HE COUNTER CASH

—

opening + Σ cash – Σ out

PAYTM WALLET

—

awaiting CSV upload

OPEN CAPTAIN OWES

—

PINs with non-zero

EQUATIONS · THE MATHEMATICAL SPINE

```
EQ_HE_CAPTAIN_CASH_OWED(USR_PIN_X) =
  Σ pos.payment WHERE
    config_id ∈ {POS_5, POS_6, POS_32}
    AND pm_id ∈ {PM_11, PM_19, PM_64}
    AND user_id_pin = PIN_X
  ⇒ MUST = cash returned at shift close per PIN
```

```
EQ_HE_PAYTM_RECON =
  Σ pos.payment WHERE pm_id ∈ {PM_14, PM_52, PM_58}
  vs Paytm CSV (manual upload pending API key)
  ⇒ drift > ₹500 fires RULE_H_PAYTM_DRIFT
```

```
EQ_HE_AGG_NET_EXPECTED(platform) =
  Σ POS_7_gross × (1 - comm_pct ± 0.5%)
  ⇒ match BNK_HDFC_HE credit T + contract days
```

```
EQ_HE_CASH_DRAWER =
  opening_float
  + Σ cash_in (PM_11 + captain settled + waiter settled)
  - Σ cash_out (counter expenses + handovers_out)
  ⇒ MUST = physical drawer count
```

```
EQ_LIVE_INVENTORY(item) = // shared with NCH
  closing_count(yesterday)
  + Σ purchase.receive(item, today)
  - Σ pos.sale × BoM(item, product)
  - Σ wastage(item, today)
```

CLAUDE AGENT LAYER · HE-SPECIFIC FUTURE

AG_HE_INVENTORY_WATCHDOG

AG_HE_PAYTM_CSV_RECON manual

AG_HE_CAPTAIN_CLOSURE

AG_HE_AGG_RECON

AG_HE_CASH_RECONCILE

AG_HE_COMP_AUDIT

ASSET CODE REGISTRY · HE

CODE	NAME	ACTIVE
P0S_5	HE - Cash Counter	✓ central settle
P0S_6	HE - Captain	✓ field collector
P0S_32	HE - GF Waiter Non-AC	✓ field collector
P0S_7	HE - Delivery	✓ scraper-fed
P0S_10	HE - WABA	baseline · not active

CODE	NAME	JOURNAL
PM_11	HE Cash Counter · is_cash	HE - Cash Counter
PM_12	HE Card · low use	Razorpay
PM_14	HE UPI Counter	Razorpay → Paytm now
PM_15	HE Swiggy	Swiggy
PM_16	HE Zomato	Zomato
PM_17	HE WABA · baseline	HE WABA UPI
PM_19	HE Cash Captain · is_cash	HE - Cash Captain
PM_52	HE UPI Captain	HE Captain 01 UPI
PM_57	HE Complimentary	HE Complimentary
PM_58	HE UPI · shared	HDFC Bank - HN Hotels
PM_64	HE Cash GF Waiter · is_cash	HE - Cash GF Waiter
PM_65	HE Complimentary (waiter)	HE Complimentary

CODE	SCOPE	NOTE
USR_PIN_*	any active hr_employee with HE-counter access	flexible · whoever logs in owns settlement
USR_PIN_CASHIER	person at HE counter station	collects from captain/waiter at shift close
USR_PIN_CAPTAIN	field collector at AC floor	1 active today · flexible
USR_PIN_WAITER	field collector at GF Non-AC	1 active today · flexible

CODE	PURPOSE	STATE
PAYTM_HE	UPI gateway for PM_14/52/58	no API key — CSV recon
BNK_HDFC_HE	Paytm payout T+1 · aggregator settlement	✓ active
AGG_SWIGGY	commission 24.60% ± 0.5%	contract 02 Mar 2026
AGG_ZOMATO_DEL	commission 23.41% ± 0.5%	contract 02 Mar 2026
AGG_ZOMATO_DINING	commission 8.26%	HE only · valid till 2046
AGG_EZD	EazyDiner ₹30/cover + 5% + 1.8%	HE only · 28 Mar 2026

CODE	ROUTE	ROLE
DEP_HE_V2	hamzaexpress.in/ops/v2	Cashier UI · drives captain/waiter cash settlement
DEP_HE_SALES	/ops/sales	HE sales analytics
DEP_HE_ITEM_COUNTER	/ops/item-counter	Live item counter (existing)
DEP_HE_KITCHEN	/ops/kitchen-intel	Prep times · KDS
DEP_AGG	/ops/aggregator (HQ)	Shared scraper for Swiggy + Zomato
DEP_HQ_BANK	/ops/bank (HQ)	Shared HDFC ledger

CODE	ROUTE	ROLE
DEP_HQ_MONEY	/ops/money (HQ)	Shared cash + bank cockpit

NCH ↔ HE · SIDE-BY-SIDE

ASPECT	NCH	HE
Odoo instance	ops.hamzahotel.com · C0_10	test.hamzahotel.com · C0_1
POS configs	POS_27 · 28 · 29	POS_5 · 6 · 7 · 10 · 32
Cash actor identity	Runner with barcode (RNR_64–68)	POS-user PIN (flexible, any employee)
Cash actor count	5 fixed runners	1 captain + 1 waiter today (flexible)
Promise mechanic	Cash AND UPI (per-runner Razorpay QR)	Cash only · UPI direct to bank
UPI gateway	Razorpay live API + webhook	Paytm CSV (no API key yet) · manual recon
QR fan-in	5 dedicated runner QRs + 2 counter QRs	None per-captain · single Paytm gateway
Token Issuance	PM_48 (chai tokens to runners)	n/a
Aggregator source	POS_29 dead · /ops/aggregator scrape	POS_7 · /ops/aggregator scrape (same)
Commission rates	Swiggy 24.60% · Zomato Del 23.41%	Same + Zomato Dining 8.26% · EZD ₹30+5%+1.8%
Bank pile	BNK_HDFC_MAIN + BNK_HDFC_OPS (split)	BNK_HDFC_HE (same HDFC currently)
Complementary	PM_49 → D3 named-owes	PM_57 + PM_65 → D3 named-owes (same pattern)
Avg ticket	~₹349 (chai cafe)	~₹464 (QSR biryani)

NEXT · NOW BOTH BRANDS HAVE L1 LOCKED

L1 · INFLOW ✓

L2 · OUTFLOW

L3 · DAILY SETTLEMENT

NCH-L1